

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 17: Special Types of Quadrilaterals

Topic-wise short notes with solved examples

The trapezium, parallelogram, rectangle, rhombus, square and kite: their definitions, properties of sides, angles and diagonals, and problems based on them.

## 1 Quadrilaterals

- A **quadrilateral** has 4 sides, 4 angles, 4 vertices and 2 diagonals.
- The **sum of its angles is**  $360^\circ$ .
- **Adjacent sides** meet at a vertex; **opposite sides** do not. **Adjacent angles** share a side; **opposite angles** do not.

**Example 1.1** The angles of a quadrilateral are in the ratio 3 : 4 : 5 : 6. Find them.

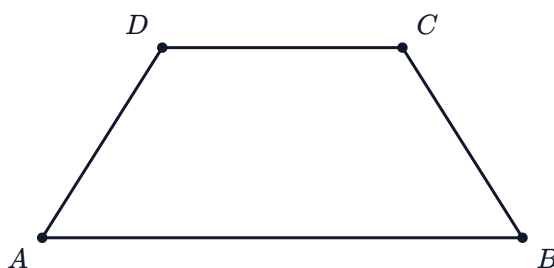
SOLUTION

$$18x = 360^\circ, \text{ so } x = 20^\circ$$

$$60^\circ, 80^\circ, 100^\circ, 120^\circ$$

## 2 Trapezium

- A **trapezium** has **one pair of opposite sides parallel**.
- The two angles on each non-parallel side add up to  $180^\circ$  (co-interior angles).
- An **isosceles trapezium** has equal non-parallel sides. Its base angles are equal and its diagonals are equal.



Isosceles trapezium:  $AB \parallel DC$  and  $AD = BC$

**Example 2.1** In trapezium ABCD,  $AB \parallel DC$ ,  $\angle A = 65^\circ$  and  $\angle B = 80^\circ$ . Find  $\angle C$  and  $\angle D$ .

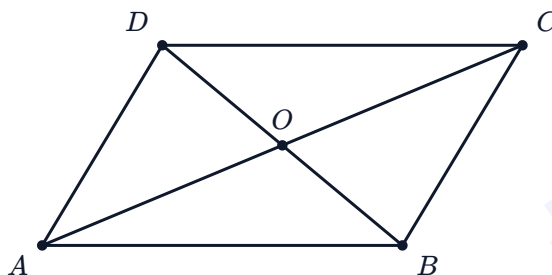
SOLUTION

$$\angle D = 180^\circ - 65^\circ \text{ and } \angle C = 180^\circ - 80^\circ$$

$$\angle C = 100^\circ, \angle D = 115^\circ$$

### 3 Parallelogram

- A **parallelogram** has **both pairs of opposite sides parallel**.
- Opposite sides are **equal**.
- Opposite angles are **equal**.
- Adjacent angles are **supplementary** (add up to  $180^\circ$ ).
- The diagonals **bisect each other**.
- Each diagonal divides it into two congruent triangles.



Parallelogram:  $OA = OC$  and  $OB = OD$

**Example 3.1** In parallelogram ABCD,  $\angle A = 70^\circ$ . Find the other angles.

SOLUTION

$$\angle C = 70^\circ, \angle B = \angle D = 110^\circ$$

**Example 3.2** Two adjacent sides of a parallelogram are in the ratio 3 : 5 and its perimeter is 64 cm. Find its sides.

SOLUTION

$$2(3x + 5x) = 64, \text{ so } x = 4$$

**12 cm and 20 cm**

**Example 3.3** In parallelogram PQRS,  $\angle P = (2x + 15)^\circ$  and  $\angle R = (3x - 20)^\circ$ . Find  $x$  and all the angles.

SOLUTION

Opposite angles are equal:  $2x + 15 = 3x - 20$ , so  $x = 35$ .

$$\angle P = \angle R = 85^\circ, \angle Q = \angle S = 95^\circ$$

### 4 Rectangle

- A **rectangle** is a parallelogram in which **every angle is  $90^\circ$** .
- It has all the properties of a parallelogram.
- Its diagonals are **equal** and bisect each other, so the four half-diagonals are equal.

**Example 4.1** The diagonals of rectangle ABCD meet at O. If  $\angle AOB = 70^\circ$ , find  $\angle OAB$  and  $\angle OBC$ .

SOLUTION

$OA = OB$ , so  $\angle OAB = \frac{180^\circ - 70^\circ}{2} = 55^\circ = \angle OBA$ . Then  $\angle OBC = 90^\circ - 55^\circ$ .

**$55^\circ$  and  $35^\circ$**

**Example 4.2** A rectangle has sides 9 cm and 12 cm. Find the length of a diagonal.

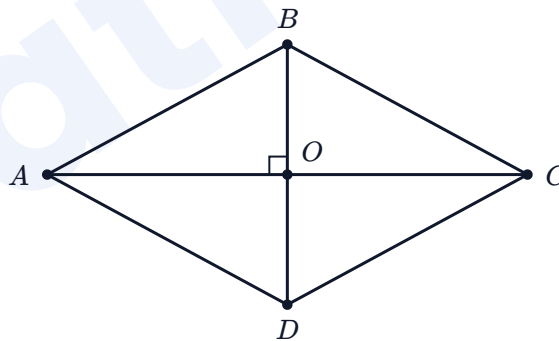
SOLUTION

$$\sqrt{81 + 144}$$

**15 cm**

## 5 Rhombus

- A **rhombus** is a parallelogram with **all sides equal**.
- It has all the properties of a parallelogram.
- Its diagonals **bisect each other at right angles**.
- Each diagonal **bisects the angles** at the vertices it joins.
- The diagonals are not equal (unless it is a square).



Rhombus: the diagonals bisect each other at right angles

**Example 5.1** The diagonals of a rhombus are 16 cm and 12 cm. Find its side and perimeter.

SOLUTION

Half-diagonals 8 and 6: side =  $\sqrt{64 + 36} = 10$  cm

**10 cm; 40 cm**

**Example 5.2** In rhombus ABCD,  $\angle ABC = 56^\circ$ . Find  $\angle ACD$ .

SOLUTION

$\angle BCD = 180^\circ - 56^\circ = 124^\circ$ . Diagonal AC bisects it.

**$62^\circ$**

## 6 Square

- A **square** is a rectangle with all sides equal; it is also a rhombus with every angle  $90^\circ$ .
- All sides are equal and every angle is  $90^\circ$ .
- The diagonals are **equal, bisect each other at right angles**, and make  $45^\circ$  with the sides.
- Diagonal of a square of side  $a$  is  $a\sqrt{2}$ .

**Example 6.1** The diagonal of a square is  $7\sqrt{2}$  cm. Find its side and area.

SOLUTION

**7 cm; 49 sq cm**

## 7 Kite

- A **kite** has **two pairs of equal adjacent sides**:  $AB = AD$  and  $CB = CD$ .
- The diagonal AC (the axis of symmetry) is the **perpendicular bisector** of BD.
- AC bisects  $\angle A$  and  $\angle C$ . The angles at B and D are equal.

**Example 7.1** In kite ABCD with  $AB = AD$  and  $CB = CD$ ,  $\angle A = 100^\circ$  and  $\angle C = 60^\circ$ . Find  $\angle B$ .

SOLUTION

$\angle B = \angle D$ , so  $2\angle B = 360^\circ - 160^\circ$   
 **$100^\circ$**

## 8 Comparing the diagonals

| Property of the diagonals | Parallelogram | Rectangle | Rhombus | Square | Kite     |
|---------------------------|---------------|-----------|---------|--------|----------|
| Bisect each other         | Yes           | Yes       | Yes     | Yes    | No       |
| Equal                     | No            | Yes       | No      | Yes    | No       |
| Perpendicular             | No            | No        | Yes     | Yes    | Yes      |
| Bisect the angles         | No            | No        | Yes     | Yes    | One only |

- Every square is a rectangle and a rhombus. Every rectangle and every rhombus is a parallelogram. Every parallelogram is a trapezium in the sense of having a pair of parallel sides.

## 9 Common mistakes

- Taking adjacent angles of a parallelogram as equal instead of supplementary.
- Saying the diagonals of a parallelogram are equal.
- Saying the diagonals of a rectangle are perpendicular.
- Using the full diagonals instead of half-diagonals to find the side of a rhombus.
- Forgetting that the diagonals of a rhombus bisect its angles.
- Treating a kite as a parallelogram.

MathSetu